UNIVERSIDADE DE SÃO PAULO

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Bayesian inference for nonhomogeneous Poisson processes in software reliability models assuming nonmonotonic intensity functions.

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ABSTRACT "In this paper, we introduce a Bayesian analysis for nonhomogeneous Poisson process in software reliability models assuming nonmonotonic intensity functions. Posterior summaries of interest are obtained using Markov Chain Monte Carlo Methods. We also present a Bayesian criterium to discriminate different models for the intensity function assuming a bathtub form, and we illustrate the proposed methodology with a numerical example".

Key Words: Superposition of Poisson processes, Bayesian analysis, exponentiated-Weibull intensity function, incidence probability, marginal likelihood.

1 Introduction

The modeling of the number of failures of a software could be based on a point process to count failures (see for example, Singpurwalla and Wilson, 1994; or Musa, Iannino and Okumoto, 1987). Let $M(t)$ be the cumulative number of failures of the software that are observed during time $(0,t]$ and $M(t)$ is modeled by a nonhomogeneous Poisson process (NHPP) with mean value function $m(t)$. The cumulative number of failures $M(t)$ can also be specified by its intensity function $\lambda(t)$ which is the derivative of $m(t)$ with respect to t; either of these functions completely specify a particular NHPP, with distribution,

$$P[M(t)=n]=\frac{[m(t)]^n}{n!}e^{-m(t)} \tag{1}$$

where $n=0,1,2,\cdots$

When we observe that there is an unknown number N of faults at the beginning of software debugging and we model the epochs of failures to be the first n order statistics taken from N i.i.d. observations with density f and c.d.f. F (a general order statistics model), we can consider the mean value function given by

$$m(t) = \xi F(t) \tag{2}$$

Observe that in this case, $\lim_{t\to\infty} m(t)$ is finite and this process can be denoted by NHPP-I.

From (2), we observe that if $F(t) = 1 - e^{-\beta t}$, we have the Goel and Okumoto (1979) process (an exponential order statistics model); if $F(t) = 1 - e^{-\beta t^{\alpha}}$, we have the Goel (1983) process (a Weibull order statistics model); if $F(t) = 1 - (1 + \beta t)e^{-\beta t}$, we have the Ohba-Yamada process (see Yamada, 1984); If $F(t) = I_k(\beta t^{\alpha})$, where $I_k(s)$ is the incomplete gamma integral given by

$$I_k(s) = \frac{1}{\Gamma(k)} \int_0^s x^{k-1} e^{-x} dx$$

we have a generalized gamma order statistics model (see Achcar, Dey and Niverthy, 1998).

Other NHPP are also proposed to model software reliability data. In situations where new faults are introduced during debugging, we need to replace the general order statistics (GOS) model with a record value statistics model (RVS). In this case, we have $m(t) \to \infty$ *as* $t \to \infty$, and we denote such as NHPP-II (see Yang, 1994). A special case is given by

$$m(t) = -ln(1 - F(t)) \tag{3}$$

which reduces to the cases of the Musa-Okumoto (1984) process with $\lambda(t) = \alpha/(t + \beta)$, the Duane (1964) process with $\lambda(t) = \alpha\beta t^{\alpha-1}$ and Cox and Lewis (1966) process with $\lambda(t) = exp(\alpha + \beta t)$.

When the intensity function is assumed to be a nonmonotonic function, we can consider the superposition of several independent NHPP with simple intensity functions (see for example, Kuo and Yang, 1996). The points of failure of a superposition process are defined to be the union of the points of failure from several component point processes.

Kuo and Yang (1996) introduce a Bayesian analysis for NHPP-I software reliability models considering special choices of F(t) in (2) using Gibbs sampling (see for example, Gelfand and Smith, 1990) with Metropolis-Hastings algorithms (see for example, Chib and Greenberg, 1995; or Smith and Roberts, 1993). They also introduce Bayesian inference using MCMC (Markov Chain Monte Carlo) methods for the superposition of NHPP.

In this paper, we consider the use of MCMC methods to obtain Bayesian inferences for a special case of NHPP-II: the use of exponentiated-Weibull form (see Mudholkar et al., 1995) for the intensity function $\lambda(t)$. With this choice for $\lambda(t)$, we can have nonmonotonic intensity functions. We also introduce an alternative to the choice of exponentiated-Weibull intensity functions, considering the superposition of NHPP, and some Bayesian criteria to discriminate the proposed models for a data set.

2 A NHPP-II assuming exponentiated-Weibull intensity functions

Consider the exponentiated-Weibull form for the intensity function $\lambda(t)$ (see for example, Mudholkar et al., 1995), given by,

$$\lambda(t) = \frac{\alpha\theta[1 - exp(-(t/\sigma)^{\alpha})]^{\theta-1} exp(-(t/\sigma)^{\alpha})(t/\sigma)^{\alpha-1}}{\sigma(1 - [1 - exp(-(t/\sigma)^{\alpha})]^{\theta})} \tag{4}$$

The great flexibility of this model to fit software reliability data is given by the different forms that the intensity function (4) can take, that is,

(i) If $\alpha \geq 1$ *and* $\alpha\theta \geq 1$, we have a monotone increasing intensity function;

(ii) If $\alpha \leq 1$ *and* $\alpha\theta \leq 1$, we have a monotone decreasing intensity function;

(iii)If $\alpha > 1$ *and* $\alpha\theta < 1$, we have a bathtub form for the intensity function, that is $0 < \theta < 1$;

(iv) If $\alpha < 1$ *and* $\alpha\sigma > 1$, we have a unimodal intensity function;

The mean value function $m(t)$ is given by (3), where

$$F(t) = [1 - exp(-(t/\sigma)^{\alpha})]^{\theta} \tag{5}$$

Assume the failure truncated model, where the ordered epochs of the observed n failures are denoted by $t_1, t_2, \cdots, t_n$.

The likelihood function (see for example, Cox and Lewis, 1966; or Lawless, 1982) is given by

$$L(\alpha, \theta, \sigma|\mathcal{D}) = (\prod_{1}^{n} \lambda(t_i)) exp(-m(t_n)) \tag{6}$$

where $\mathcal{D}$ denoted the data set, $\lambda(t)$ *and* $m(t)$ are given by (3) and (4), respectively.

That is,

$$L(\alpha, \theta, \sigma|\mathcal{D}) = \frac{\alpha^n \theta^n A(\alpha, \theta, \sigma) S(t_n)}{\sigma^{n\alpha} \prod_1^n S(t_i)} \tag{7}$$

where $S(t_i) = 1 - F(t_i) = 1 - (1 - exp(-(t_i/\sigma)^{\alpha})))^{\theta}$

and $A(\alpha, \theta, \sigma) = \prod_1^n t_i^{\alpha-1} exp(-(t_i/\sigma)^{\alpha})(1 - exp(-(t_i/\sigma)^{\alpha}))^{\theta-1}$.

For Bayesian inference, we assume the following prior densities for α, θ *and* σ:

$$\begin{array}{ll} \text{(i)} & \alpha \sim \Gamma(a_1, b_1) \\ \text{(ii)} & \theta \sim \Gamma(a_2, b_2) \\ \text{(iii)} & \sigma \sim \Gamma(a_3, b_3) \end{array} \tag{8}$$

where a_1, a_2, a_3, b_1, b_2 *and* b_3 are known and $\Gamma(a, b)$ denotes a gamma distribution with mean a/b and variance a/b^2. We further assume independence among the parameters α, θ *and* σ.

The joint posterior density is,

$$\pi(\alpha,\theta,\sigma|\mathcal{D}) \propto \alpha^{a_1+n-1}\theta^{a_2+n-1}\sigma^{a_3-n\alpha-1}exp(-b_1\alpha-b_2\theta-b_3\sigma)\frac{A(\alpha,\theta,\sigma)b(\alpha,\theta,\sigma)}{B(\alpha,\theta,\sigma)} \tag{9}$$

where $b(\alpha,\theta,\sigma)=S(t_n)$, $B(\alpha,\theta,\sigma)=\prod_1^n S(t_i)$ *and* $S(t_i)$ is defined in (7).

The conditional posterior densities for the Gibbs sampling algorithm are given by

$$\begin{aligned}
&\text{(i)} && \pi(\alpha|\theta,\sigma,\mathcal{D}) \propto \alpha^{a_1-1}e^{-b_1\alpha}\psi_1(\alpha,\theta,\sigma)\\
&\text{where} && \psi_1(\alpha,\theta,\sigma)=exp(n\ln\alpha+\ln A(\alpha,\theta,\sigma)+\ln b(\alpha,\theta,\sigma)-\ln B(\alpha,\theta,\sigma))\\
&\text{(ii)} && \pi(\theta|\alpha,\sigma,\mathcal{D}) \propto \theta^{a_2-1}e^{-b_2\theta}\psi_2(\alpha,\theta,\sigma)\\
&\text{where} && \psi_2(\alpha,\theta,\sigma)=exp(n\ln\theta+\ln A(\alpha,\theta,\sigma)+\ln b(\alpha,\theta,\sigma)-\ln B(\alpha,\theta,\sigma))\\
&\text{(iii)} && \pi(\sigma|\alpha,\theta,\mathcal{D}) \propto \sigma^{a_3-1}e^{-b_3\sigma}\psi_3(\alpha,\theta,\sigma)\\
&\text{where} && \psi_3(\alpha,\theta,\sigma)=exp(-n\alpha\ln\sigma+\ln A(\alpha,\theta,\sigma)+\ln b(\alpha,\theta,\sigma)-\ln B(\alpha,\theta,\sigma))
\end{aligned} \tag{10}$$

The variables α,θ *and* σ should be generated using the Metropolis-Hastings algorithm.

3 Superposition of NHPP

Other possibility to model software reliability data when the intensity functions are not monotonic, is to consider the superposition of several independent NHPP with simple intensity functions.

Let $M_j(t)$ denote the NHPP for the failures from the j^{th} component in $(0,t]$, with intensity function $\lambda_j(t|\underline{\beta_j})$, where the function form of $\lambda_j(t|\underline{\beta_j})$ is assumed to be known with unknown parameter $\underline{\beta_j}$ that may be a vector. Also assume that the number of failures from the j^{th} component $M_j(t)$, $j=1,\cdots,J$ are independent. A process $M(t)=\sum_{j=1}^J M_j(t)$ which counts the number of failures in the interval $(0,t]$ for the superposition model is also a NHPP with intensity function,

$$\lambda(t|\underline{\beta})=\sum_{j=1}^{J}\lambda_j(t|\underline{\beta_j}) \tag{11}$$

where $\underline{\beta}=(\underline{\beta_1},\underline{\beta_2},\cdots,\underline{\beta_J})$.

With superposition of NHPP, we can have many different forms for the intensity functions: bathtub-shaped fuctions, polynomials with peaks and valleys, and simple monotonic functions.

A special case (see Musa and Okumoto,1984) is given by

$$\lambda(t|\underline{\beta})=\frac{\alpha}{\beta_1+t}+\beta_2 t \tag{12}$$

Assuming the superposition of NHPP with intensity function (12) and the failure truncated model, the likelihood function (see (6)) for α, β_1 *and* β_2 is given by

$$L(\alpha,\beta_1,\beta_2) = (\prod_{i=1}^{n}(\frac{\alpha}{\beta_1+t_i}+\beta_2 t_i))exp(-\alpha ln(1+\frac{t_n}{\beta_1})-\frac{\beta_2 t_n^2}{2}) \tag{13}$$

Assume the following prior distributions for α, β_1 *and* β_2 :

$$\begin{array}{ll} \text{(i)} & \alpha \sim \Gamma(a_1,b_1) \\ \text{(ii)} & \beta_1 \sim \Gamma(a_2,b_2) \\ \text{(iii)} & \beta_2 \sim \Gamma(a_3,b_3) \end{array} \tag{14}$$

where a_1, a_2, a_3, b_1, b_2 *and* b_3 are known. Assuming prior independence among the parameters, the joint posterior distribution for α, β_1, β_2 is given by

$$\pi(\alpha,\beta_1,\beta_2|\mathcal{D}) \propto (\prod_{i=1}^{n}(\frac{\alpha}{\beta_1+t_i}+\beta_2 t_i))\alpha^{a_1-1}\beta_1^{a_2-1}\beta_2^{a_3-1}e^{-b_2\beta_1}exp(-(b_1+ln(1+\frac{t_n}{\beta_1}))\alpha-(b_3+\frac{t_n^2}{2})\beta_2) \tag{15}$$

Consider the introduction of latent variables (see Tanner and Wong, 1987), $\underline{I_i} = (I_{i1}, I_{i2})$ where $I_{ij} = 1$ if the i^{th} failure is caused by the j^{th} type and $I_{ij} = 0$ otherwise, $i = 1, \cdots, n$.

Observe that $I_{i1} + I_{i2} = 1$, for $i = 1, \cdots, n$.

Assume that the conditional distribution for $\underline{I_i}$ given $\underline{\beta}$ and $\mathcal{D}$ is a Bernoulli distribution with sucess probability p_{i1} given by

$$p_{i1} = \frac{\alpha/(\beta_1+t)}{\alpha/(\beta_1+t)+\beta_2 t} \tag{16}$$

That is,

$$\begin{aligned} \pi(\underline{I_1},\underline{I_2},\cdots,\underline{I_n}|\beta,\mathcal{D}) &\propto \prod_{i=1}^{n} p_{i1}^{I_{i1}}(1-p_{i1})^{I_{i2}} \\ &\propto \frac{\alpha^{\sum_{i=1}^{n} I_{i1}}\beta_2^{\sum_{i=1}^{n} I_{i2}}(\prod_{i=1}^{n} t_i^{I_{i2}})}{(\prod_{i:I_{i1}=1}^{n}(\beta_1+t_i))(\prod_{i=1}^{n}(\frac{\alpha}{\beta_1+t_i}+\beta_2 t_i))} \end{aligned} \tag{17}$$

In this way, the posterior density for α, β_1 and β_2 given $\underline{I_1}, \underline{I_2}, \cdots, \underline{I_n}$ and $\mathcal{D}$ is given by

$$\begin{aligned} \pi(\alpha,\beta_1,\beta_2|\underline{I_1},\underline{I_2},\cdots,\underline{I_n},\mathcal{D}) \propto\ & \frac{1}{\prod_{i:I_{i1}=1}^{n}(\beta_1+t_i)} \\ & \alpha^{a_1-1+\sum_{i=1}^{n} I_{i1}}\beta_2^{a_3-1+\sum_{i=1}^{n} I_{i2}}\beta_1^{a_2-1}e^{-b_2\beta_1} \\ & exp(-(b_1+\ln(1+\frac{t_n}{\beta_1}))\alpha-(b_3+\frac{t_n^2}{2})\beta_2) \end{aligned} \tag{18}$$

The condititional distributions for the Gibbs algorithm are given by

(i) Construct $I = (\underline{I_1}, \underline{I_2}, \cdots, \underline{I_n})^T$ (a nx2 matrix) given β and $\mathcal{D}$ generating independent variables I_{i1} from the Bernoulli distribution with parameter

$$p_{i1} = \frac{\alpha}{\alpha+\beta_2(\beta_1+t_i)t_i}$$

(ii)

$$\pi(\alpha|\beta_1,\beta_2,I,\mathcal{D}) \sim \Gamma\Big(a_1+\sum_{i=1}^{n} I_{i1}; b_1+\ln\Big(1+\frac{t_n}{\beta_1}\Big)\Big)$$

(iii)

$$\pi(\beta_2|\alpha,\beta_1,I,\mathcal{D}) \sim \Gamma\Big(a_3+n-\sum_{i=1}^{n} I_{i1}; b_3+\frac{t_n^2}{2}\Big) \tag{19}$$

(iv)

$$\pi(\beta_1|\alpha,\beta_2,I,\mathcal{D}) \propto \beta_1^{a_2-1}e^{-b_2\beta_1}\psi(\alpha,\beta_1,\beta_2)$$

$$\text{where} \quad \psi(\alpha,\beta_1,\beta_2) = exp\Big(-\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-\sum_{i=1}^{n} I_{i1}\ln\Big(\beta_1+t_i\Big)\Big)$$

Observe that, we need to use the Metropolis algorithm to generate the variable β_1.

4 Superposition of NHPP considering the introduction of incidence probabilities

Assume now, the superposition of NHPP model with intensity function

$$\lambda(t|\underline{\theta},\underline{\beta}) = \sum_{j=1}^{J}\theta_j\lambda_j(t|\underline{\beta_j}) \tag{20}$$

where the incidence probabilities θ_j satisfies $\sum_{j=1}^{J}\theta_j = 1$ and $\underline{\theta} = (\theta_1,\theta_2,\cdots,\theta_J)$.

A special case of (20) is the generalization of the Musa and Okumoto (1984) intensity function given by

$$\lambda(t|\theta,\underline{\beta}) = \frac{\theta\alpha}{\beta_1+t}+(1-\theta)\beta_2 t \tag{21}$$

where $\theta_1 = \theta$ when $J = 2$.

In this case, the mean value function $m(t)$ is given by

$$m(t|\underline{\theta},\underline{\beta}) = \theta\alpha\ln(1+\frac{t}{\beta_1})+(1-\theta)\beta_2\frac{t^2}{2} \tag{22}$$

Assuming the failure truncated model, the likelihood function for $\theta,\alpha,\beta_1,$ and β_2 is (from (6), (21), and (22)) given by

$$L(\theta,\alpha,\beta_1,\beta_2|\mathcal{D}) = \Big(\prod_{i=1}^{n}\Big(\frac{\alpha\theta}{\beta_1+t_i}+(1-\theta)\beta_2 t_i\Big)\Big)exp\Big(-\theta\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-(1-\theta)\frac{\beta_2 t_n^2}{2}\Big) \tag{23}$$

Assuming the same prior distributions (14) for $\alpha, \beta_1,$ and β_2 and a Beta $B(a_4, b_4)$ distribution for θ with a_4 and b_4 known, and prior independence among the parameters, the joint posterior distribution for α, β_1, β_2 and θ is given by

$$\pi(\alpha,\beta_1,\beta_2,\theta|\mathcal{D}) \propto \Big(\prod_{i=1}^{n}\Big(\frac{\alpha\theta}{\beta_1+t_i}+(1-\theta)\beta_2 t_i\Big)\Big)\alpha_1^{a_1-1}\beta_1^{a_2-1}\beta_2^{a_3-1}\theta^{a_4-1}(1-\theta)^{b_4-1}$$
$$exp\Big(-b_1\alpha-b_2\beta_1-b_3\beta_2-\theta\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-(1-\theta)\frac{\beta_2 t_n^2}{2}\Big) \tag{24}$$

Also considering the introduction of latent variables $\underline{I_i}=(I_{i1},I_{i2})$, where $I_{ij}=1$ if the i^{th} failure is caused by the j^{th} type and $I_{ij}=0$ otherwise $i=1,\cdots,n$ with a Bernoulli distribution with parameter

$$p_{i1}=\frac{\theta\alpha/(\beta_1+t_i)}{\theta\alpha/(\beta_1+t_i)+(1-\theta)\beta_2 t_i} \tag{25}$$

we have

$$\pi(\underline{I_1},\cdots,\underline{I_n}|\underline{\beta},\underline{\theta},\mathcal{D}) \propto \frac{(\theta\alpha)^{\sum_{i=1}^{n}I_{i1}}\Big((1-\theta)\beta_2\Big)^{\sum_{i=1}^{n}I_{i2}}\Big(\prod_{i=1}^{n}t_i^{I_{i2}}\Big)}{\Big(\prod_{i:I_{i1}=1}^{n}(\beta_1+t_i)\Big)\Big(\prod_{i=1}^{n}\Big(\theta\alpha/(\beta_1+t_i)+(1-\theta)\beta_2 t_i\Big)} \tag{26}$$

where $I_{i1}+I_{i2}=1$.

The joint posterior distribution for $\alpha,\beta_1,\beta_2,$ and θ given $I=(\underline{I_1},\cdots,\underline{I_n})$ and $\mathcal{D}$ is given by

$$\begin{aligned}\pi(\alpha,\beta_1,\beta_2,\theta|I,\mathcal{D}) \quad \propto \quad & \frac{1}{\Big(\prod_{i:I_{i1}=1}^{n}(\beta_1+t_i)\Big)}\alpha^{a_1-1+\sum_{i=1}^{n}I_{i1}}\beta_1^{a_2-1}\beta_2^{a_3-1+\sum_{i=1}^{n}I_{i2}}\theta^{a_4-1+\sum_{i=1}^{n}I_{i1}}\\ & (1-\theta)^{b_4-1+\sum_{i=1}^{n}I_{i2}}e^{-b_1\alpha}e^{-b_2\beta_1}e^{-b_3\beta_2}\\ & exp\Big(-\theta\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-(1-\theta)\frac{\beta_2 t_n^2}{2}\Big)\end{aligned} \tag{27}$$

The conditional distributions for the Gibbs algorithm are given by

(i) Construct $I=(\underline{I_1},\cdots,\underline{I_n})^T$ (a nx2 matrix) given $\underline{\beta}$ and $\mathcal{D}$ generating independent variables I_{i1} from the Bernoulli distribution with parameter p_{i1} given by (25).

(ii) $$\pi(\alpha|\beta_1,\beta_2,\theta,I,\mathcal{D}) \sim \Gamma\Big(a_1+\sum_{i=1}^{n}I_{i1};b_1+\theta\ln\Big(1+\frac{t_n}{\beta_1}\Big)\Big)$$

(iii) $$\pi(\beta_1|\alpha,\beta_2,\theta,I,\mathcal{D}) \propto \beta_1^{a_2-1}e^{-b_2\beta_1}\psi_1(\alpha,\beta_1,\beta_2,\theta) \tag{28}$$

$$\text{where } \psi_1(\alpha,\beta_1,\beta_2,\theta)=exp\Big(-\theta\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-\sum_{i=1}^{n}I_{i1}\ln(\beta_1+t_i)\Big)$$

(iv) $$\pi(\beta_2|\alpha,\beta_1,\theta,I,\mathcal{D}) \sim \Gamma\Big(a_3+n-\sum_{i=1}^{n}I_{i1};b_3+(1-\theta)\frac{t_n^2}{2}\Big)$$

(v) $$\pi(\theta|\alpha,\beta_1,\beta_2,I,\mathcal{D}) \propto \theta^{a_4-1+\sum_{i=1}^{n}I_{i1}}(1-\theta)^{b_4+n-1-\sum_{i=1}^{n}I_{i1}}\psi_2(\alpha,\beta_1,\beta_2,\theta)$$

$$\text{where } \psi_2(\alpha,\beta_1,\beta_2,\theta)=exp\Big(-\theta\alpha\ln\Big(1+\frac{t_n}{\beta_1}\Big)-(1-\theta)\frac{\beta_2 t_n^2}{2}\Big)$$

Observe that, we need to use the Metropolis-Hastings algorithm to generate the variables β_1 and θ

5 Bayesian inference and model determination

We can use the Gibbs samples to get inferences on the parameters of the software reliability model or on functions of these parameters. In this case, we could approximate posterior moments of interest. As a special case, consider the mean value function $m(t)$. A Bayes estimator of $m(t)$ with respect to the squared error loss function is given by the posterior mean $E(m(t)|\mathcal{D})$. Similar inferences could be obtained for $\lambda(t)$.

As a special case, consider the exponentiated-Weibull intensity function (4). Monte Carlo estimates for the posterior means of α, θ and σ could be obtained using the generated Gibbs samples. Similarly, we get an approximation for $E(\lambda(t)|\mathcal{D})$. For model selection, we could consider (see Raftery, 1996) the marginal likelihood of the whole data set $\mathcal{D}$ for a model M, given by

$$V_M = \int L(\underline{\beta}_M|\mathcal{D})\pi_M(\underline{\beta}_M)d\underline{\beta}_M \tag{29}$$

where $\underline{\beta_M}$ denotes the set of all unknown parameters in the model M, and $\pi_M(\underline{\beta}_M)$ is the prior distribution. The Bayes factor criterion prefers model M_1 to model M_2 if $V_{M_2}/V_{M_1} < 1$. A Monte Carlo estimate for the marginal likelihood V_M is given by

$$\hat{V}_M = \frac{1}{T}\sum_{t=1}^{T} L(\underline{\beta}_M^{(t)}|\mathcal{D}) \tag{30}$$

where $\underline{\beta}_M^{(t)}, t = 1, 2, \cdots, T$ could be generated using importance sampling. The simplest such estimator results from taking the prior as importance sampling function (see Raftery, 1996).

6 A numerical Illustration

The data of table 1 was simulated from a NHPP, with the exponentiated-Weibull intensity function (4) considering $\alpha = 1.2, \theta = 0.75,$ *and* $\sigma = 100$; 36 observations were generated.

i	t_i	i	t_i	i	t_i
1	1.26835	13	656.16081	25	980.52727
2	15.72966	14	671.52321	26	1062.16707
3	32.23801	15	719.57745	27	1066.97822
4	110.41090	16	780.01486	28	1092.42460
5	113.40610	17	811.17553	29	1093.95373
6	200.98991	18	826.83801	30	1136.13604
7	221.12769	19	831.33000	31	1147.00027
8	490.28306	20	886.65706	32	1340.85381
9	499.84578	21	887.42836	33	1382.16824
10	554.87830	22	917.09363	34	1410.87073
11	565.07198	23	921.82979	35	1414.14548
12	639.51697	24	968.73424	36	1560.62059

Table 1. Generated data from the Exponentiated-Weibull process with $\alpha = 1.2, \theta = 0.75, \sigma = 100$

For a Bayesian analysis of the software reliability data of Table 1, we first assume a NHPP-II process with an exponentiated-Weibull intensity function with the prior distributions for α *and* σ given in (8), where $a_1 = 7.2, b_1 = 6, a_3 = 1000, b_3 = 10$. We also consider a Beta prior distribution, $\beta(a_2, b_2)$ *for* θ *with* $a_2 = 1500$ *and* $b_2 = 500$ *since* $0 < \theta < 1$ (a bathtub form for the intensity function).

From the conditional distributions for α, θ *and* σ given in (10),we generated 10 separate Gibbs chains each of which ran for 1200 iterations. For each chain, we discarded the first 200 elements (burn-in samples). We monitored the convergence of the Gibbs samples using the Gelman and Rubin (1992) method based on the analysis of variance technique to determine if further iterations are needed. For each parameter, we considered every 10th. draw, and so we finally got a sample of size 1000.

In table 2, we have the obtained posterior summaries of the parameters α, θ *and* σ and in figure 1, we have plots of the approximate marginal densities considering S=1000 Gibbs samples. We also have in table 2, the estimated potential scale reductions $\widehat{R}$ (see Gelman and Rubin, 1992) for all parameters. In this case, the considered number of iterations were sufficient for approximate convergence ($\sqrt{\widehat{R}} < 1.1$ for all parameters).

Parameter	Mean	S.D.	95% *Credible* Interval	$\widehat{R}$
α	1.23568	0.05022	(1.12168;1.31881)	1.008
θ	0.74978	0.00911	(0.73282;0.76774)	1.007
σ	99.59496	3.20429	(93.69764;106.24767)	1.000

Table 2. Posterior summaries (exponentiated-Weibull process)

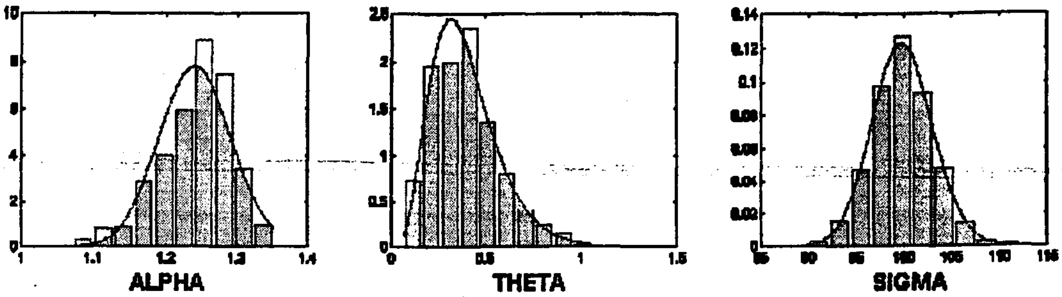

Figure 1. Approximate marginal posterior densities (exponentiated-Weibull process).

In figure 2, we observe the convergence of the Gibbs algorithm with different initial values for the parameters.

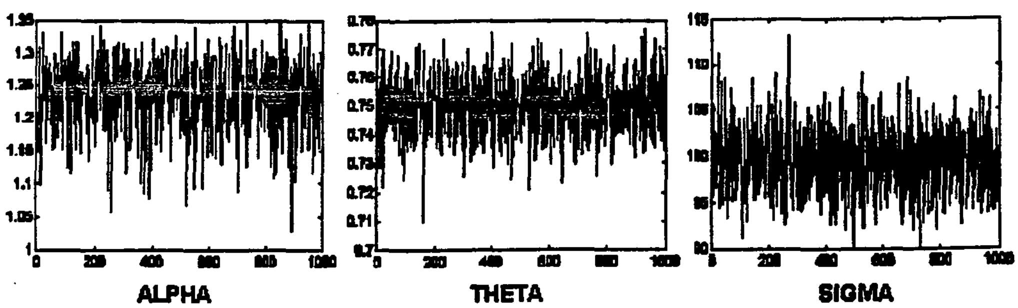

Figure 2. Convergence of the Gibbs algorithm (exponentiated-Weibull process).

Now, let us consider the superposition of the Musa-Okumoto process and an exponential process , with intensity function (12). For a Bayesian analysis consider the prior distributions (14) with $a_1 = 380, b_1 = 20, a_2 = 5000, b_2 = 50, a_3 = 0.03, b_3 = 10000$. From the conditional distributions for I, α, β_1 *and* β_2 given in (19), we generate 10 separate Gibbs chains each of which ran for 1200 iterations. We eliminated the first 200 elements of each chain, and we selected every 10th element in each chain. Thus we obtained a sample of size 1000.

In table 3, we have the obtained posterior summaries of the parameters α, β_1 *and* β_2, and in figure 3, we have the plots of the approximate marginal densities considering S=1000 Gibbs samples.

We also have in Table 3, the estimated potential scale reductions $\widehat{R}$. We observe convergence for all parameters.

Parameter	Mean	S.D.	95% *Credible* Interval	$\widehat{R}$
α	18.93163	0.90263	(17.14557;20.72774)	1.004
β_1	100.17290	2.04359	(96.00419;104.14977)	1.000
β_2	0.0000045830	0.0000035707	(0.000000017253;0.000013042)	0.998

Table 3. Posterior summaries (superposition of Musa-Okumoto and an exponential process)

We also considered the superposition of a Musa-Okumoto and an exponential process with the introduction of incidence probabilities. We assumed the prior distributions for α, β_1 *and* β_2 given by (14) with $a_1 = 500, b_1 = 20, a_2 = 2500, b_2 = 25, a_3 = 0.25, b_3 = 1000$, and for the parameter θ a Beta distribution with parameters $a_4 = 0.48$ *and* $b_4 = 0.12$. The Bayesian procedure was similar to the previous one.

In table 4, we have obtained posterior summaries of the parameters $\alpha, \beta_1, \beta_2,$ *and* θ, and in figure 4 we have the plots of the approximate marginal densities considering S=1000 Gibbs samples. We also have the estimated potencial scale reduction $\widehat{R}$ in Table 4.

Parameter	Mean	S.D.	95% *Credible* Interval	$\widehat{R}$
α	24.99587	1.10233	(22.85377;27.24914)	1.007
β_1	100.00332	2.12828	(95.76092;104.074989)	1.003
β_2	0.000023029	0.0000057885	(0.000012754;0.000035539)	1.004
θ	0.20585	0.079171	(0.073939;0.38613)	1.042

Table 4. Posterior summaries (superposition of Musa-Okumoto and an exponential process with incidence probability)

To compare the Bayes estimates of the intensity function on the simulated data of Table 1, considering the three models with $\lambda(t)$ given in (21) assuming $\alpha = 1.2, \theta = 0.75,$ *and* $\sigma = 100$, we have in Table 5, Monte Carlo estimates of $E(\lambda(t)|\mathcal{D}_t)$ based on $S = 1000$ Gibbs samples considering our three processes and the original value .

t	Original	E.W.	M.O.E. + I.P.	M.O.E.	t	Original	E.W.	M.O.E. + I.P.	M.O.E.
1.26835	0.01414	0.01300	0.05083	0.18670	831.33000	0.01833	0.02046	0.02073	0.02413
15.72966	0.01203	0.01180	0.04475	0.16346	886.65706	0.01857	0.02077	0.02143	0.02325
32.23801	0.01198	0.01197	0.03950	0.14316	887.42836	0.01857	0.02078	0.02144	0.02324
110.41090	0.01289	0.01337	0.02647	0.09041	917.09363	0.01869	0.02094	0.02183	0.02281
113.40610	0.01293	0.01343	0.02618	0.08917	921.82979	0.01871	0.02096	0.02189	0.02275
200.98991	0.01398	0.01482	0.02077	0.06379	968.73424	0.01890	0.02121	0.02253	0.02215
221.12769	0.01420	0.01511	0.02007	0.05994	980.52727	0.01894	0.02127	0.02269	0.02201
490.28306	0.01649	0.01807	0.01768	0.03430	1062.16707	0.01925	0.02167	0.02385	0.02116
499.84578	0.01656	0.01815	0.01772	0.03384	1066.97822	0.01927	0.02170	0.02392	0.02111
554.87830	0.01691	0.01860	0.01800	0.03144	1092.42460	0.01936	0.02182	0.02429	0.02088
565.07198	0.01697	0.01868	0.01807	0.03105	1093.95373	0.01925	0.02167	0.02385	0.02116
639.51697	0.01739	0.01923	0.01865	0.02853	1136.13604	0.01936	0.02183	0.02432	0.02087
656.16081	0.01748	0.01935	0.01880	0.02804	1147.00027	0.01955	0.02207	0.02510	0.02044
671.52321	0.01756	0.01945	0.01895	0.02761	1340.85381	0.02017	0.02290	0.02809	0.01928
719.57745	0.01781	0.01977	0.01944	0.02639	1382.16824	0.02029	0.02306	0.02875	0.02116
780.01486	0.018097	0.02015	0.02011	0.02508	1410.87073	0.02037	0.02317	0.02921	0.01899
811.17553	0.01824	0.02034	0.02048	0.02449	1414.14548	0.02038	0.02319	0.02926	0.01898
826.83801	0.01831	0.02043	0.02067	0.02421	1560.62059	0.02079	0.02373	0.03164	0.01855

Table 5. Original values and Monte Carlo estimates for $E(\lambda(t)|\mathcal{D})$.

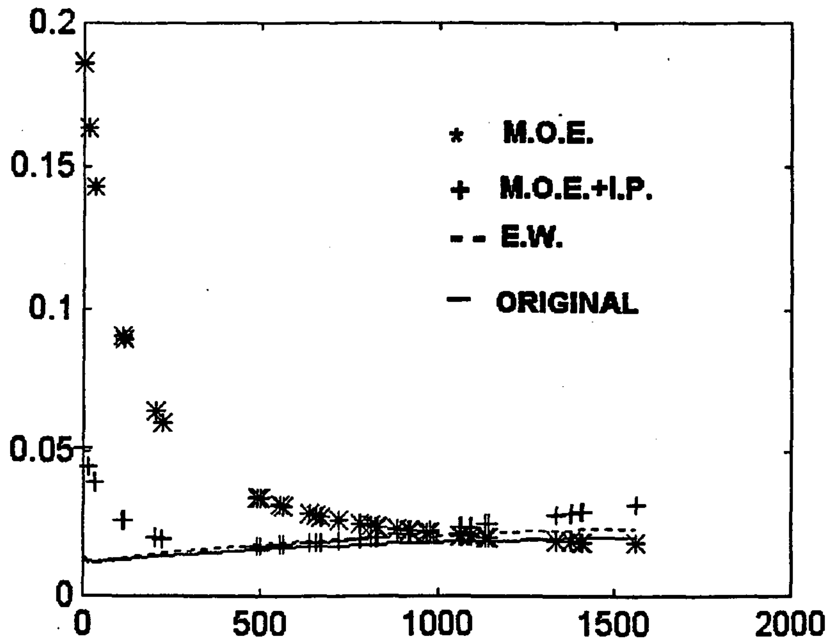

Figure 3. Comparison of the original intensity function $\lambda(t)$ and the Bayes estimates

From Table 5 and figure 3, we observe that better results are given considering the exponentiated-Weibull process (Bayes estimates close to the original values for $\lambda(t)$). Observe that M.O.E. denotes Musa-Okumoto process, M.O.E.+ I.P. denotes Musa-Okumoto process with incidence probability, E.W. denotes exponentiated-Weibull process and ORIGINAL denotes original model.

For model discrimination,we estimated the marginal likelihood from (30) for each proposed model, considering Monte Carlo estimates based on the generated Gibbs samples. These values are given in Table 6.

Model	Estimated marginal likelihood
Superposition of Musa-Okumoto and exponential	$1.450 \cdot 10^{-80}$
Superposition of Musa-Okumoto and exponential with incidence probability	$9.208 \cdot 10^{-78}$
Exponentiated-Weibull	$9.244 \cdot 10^{-77}$

Table 6. Estimated values of the marginal likelihood for model discrimination.

From Table 6, observe that the best model (large value for V_M) for the data set of Table 1 is the exponentiated-Weibull. We also can calculate from Table 6 the Bayes factors $B_{ij} = V_{M_i}/V_{M_j}, i, j = 1, 2, 3,$ for models M_1 (exponentiated-Weibull), M_2 (Superposition of Musa-Okumoto and exponential with incidence probability) and M_3 (Superposition of Musa-Okumoto and exponential). The Bayes factors are given by $B_{12} = 10.039$ (positive evidence for the model M_1), $B_{13} = 6.375 \cdot 10^3$ (very strong evidence for the model M_1) and $B_{23} = 6.3503 \cdot 10^2$ (very strong evidence for the model M_2).

7 Concluding Remarks

The use of the exponentiated-Weibull form for the intensity function is an alternative to model software reliability data since this model gives different shapes for the intensity function. Using MCMC methods, we can get accurate inference results on the parameters of the model. Other nonhomogeneous Poisson processes considering superposition of several independent NHPP also could be used to model software reliability data, especially for the case of bathtub forms for the intensity function. The Bayesian approach also gives simple methods to discriminate different models used to analyse software reliability data.

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